Applications of Derivative

Curve Sketching

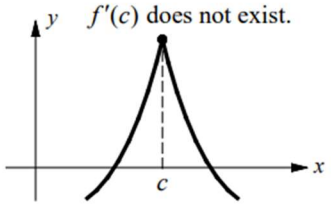
Definition of Absolute and Relative Extrema

A function f has an **absolute minimum** at c if $f(c) \le f(x)$ for all x in the domain of f.
A function f has an **absolute maximum** at c if $f(c) \ge f(x)$ for all x in the domain of f.

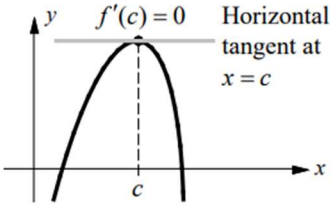
A function f has a **relative minimum** at c if $f(c) \le f(x)$ for all x in the vicinity of c.
A function f has a **relative maximum** at c if $f(c) \ge f(x)$ for all x in the vicinity of c.

Definition of a Critical Number

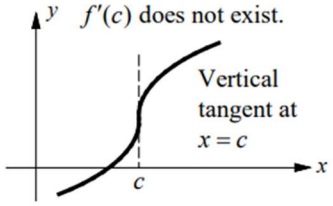
If $f'(c) = 0$ or if $f'(c)$ does not exist, then c is called a **critical number** of f.



c is a critical number of f.



c is a critical number of f.



c is a critical number of f.

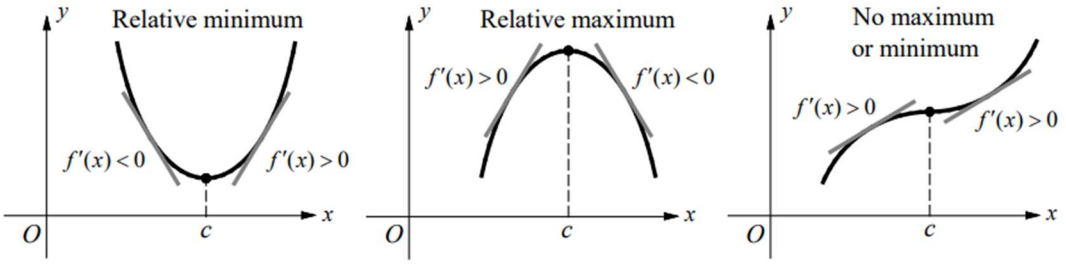
Test for Increasing and Decreasing Functions

1. A function f is **increasing** on an interval if $f'(x) > 0$ on that interval.
2. A function f is **decreasing** on an interval if $f'(x) < 0$ on that interval.

First Derivative Test

Let c be a critical number of a function f that is continuous on an open interval.

1. If $f'(x)$ changes from negative to positive at c, then $f(c)$ is a relative minimum of f.
2. If $f'(x)$ changes from positive to negative at c, then $f(c)$ is a relative maximum of f.
3. If $f'(x)$ does not change signs at c, then $f(c)$ is neither a relative minimum nor a relative maximum.



Relative Extrema Occur Only at Critical Numbers

If f has a relative minimum or a relative maximum at $x = c$, then c is a critical number of f. But the converse of this theorem is not necessarily true. The critical numbers of a function need not produce relative extrema.

The Extreme Value Theorem

If f is continuous on the closed interval $[a,b]$, then f has both a minimum and maximum on the interval.

Finding Extrema on a Closed Interval

To find the extrema of a continuous function f on a closed interval $[a,b]$, use the following steps.

1. Find the values of f at the critical numbers of f in (a,b).
2. Find the values of f at the **endpoints** of the interval.
3. The least of these values is the minimum. The greatest is the maximum.

Exercises

1. At what values of x does $f(x)=(x-1)^3(3-x)$ have the absolute maximum?

(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) $\frac{5}{2}$

2. At $x=2$, which of the following is true of the function f defined by $f(x)=x^2e^{-x}$?

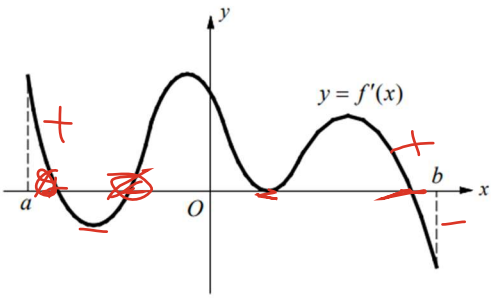
(A) f has a relative maximum.

(B) f has a relative minimum.

(C) f is increasing.

(D) f is decreasing.

3.



The graph of f', the derivative of f, is shown in the figure above. Which of the following describes all relative extrema of f on the open interval (a,b)?

(A) One relative maximum and two relative minima

(B) Two relative maxima and one relative minimum

(C) Two relative maxima and two relative minima

(D) Three relative maxima and two relative minim

Test for Concavity

1. If $f''(x) > 0$ for all x in an open interval I, then the graph of f is concave upward on I.
2. If $f''(x) < 0$ for all x in an open interval I, then the graph of f is concave downward on I.

Definition of Points of Inflection

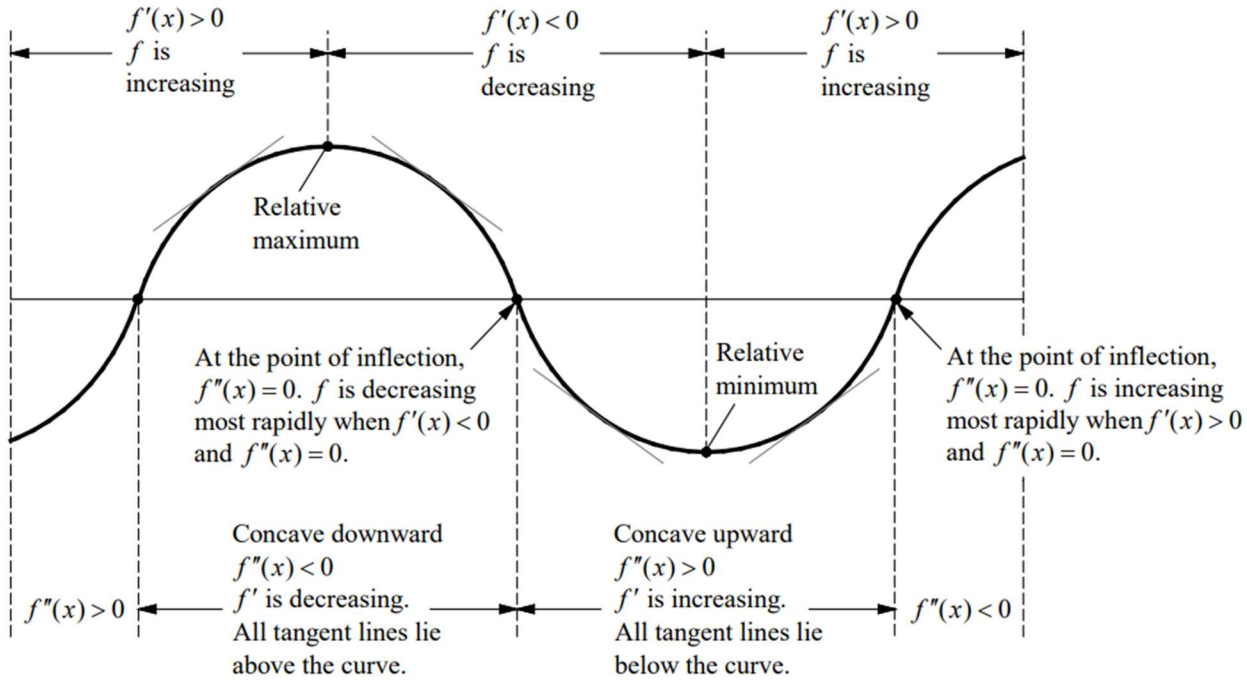
A point P on a curve $y = f(x)$ is called a point of inflection if f is continuous on P and the curve changes from concave upward to concave downward or vice versa.

If $(c, f(c))$ is a point of inflection of the graph of f, then either $f''(x) = 0$ or f is not differentiable at $x = c$.

Second Derivative Test

Let f be a function such that $f'(c) = 0$ and the second derivative of f exists on an open interval containing c.

1. If $f''(c) > 0$, then $f(c)$ is a relative minimum.
2. If $f''(c) < 0$, then $f(c)$ is a relative maximum.
3. If $f''(c) = 0$, the test fails. Use the First Derivative Test.



Exercises

1. The graph of $y = x^4 - 2x^3$ has a point of inflection at

(A) $(0,0)$ only

(B) $(0,0)$ and $(1,-1)$

(C) $(1,-1)$ only

(D) $(0,0)$ and $(\frac{3}{2},-\frac{27}{16})$

2. If the graph of $y = ax^3 - 6x^2 + bx - 4$ has a point of inflection at $(2,-2)$, what is the value of $a+b$?

(A) -2 (B) 3 (C) 6 (D) 10

3. The graph of $y = 3x^5 - 40x^3 - 21x$ is concave up for

(A) $x < 0$

(B) $x > 2$

(C) $x < 0$ or $0 < x < 2$

(D) $-2 < x < 0$ or $x > 2$

4. Let f be a twice differentiable function with $f(1) = -1$, $f'(1) = 2$, and $f''(1) = 0$. Let g be a func whose derivative is given by $g'(x) = x^2\left[2f(x) + f'(x)\right]$ for all x.

(a) Write an equation for the line tangent to the graph of f at $x = 1$.

(b) Does the graph of f have a point of inflection when $x = 1$? Explain.

(c) Given that $g(1) = 3$, write an equation for the line tangent to the graph of g at $x = 1$.

(d) Show that $g''(x) = 4xf(x) + 2x(x+1)f'(x) + x^2 f''(x)$. Does g have a local maximum or minim at $x = 1$? Explain your reasoning.

Guidelines for Sketching a Curve

1. **Domain**
   Identify the domain of f, that is, the set of values of x for which $f(x)$ is defined.

2. **Intercepts**
   To find the x-intercepts, let $y = 0$ and solve for x.
   To find the y-intercepts, let $x = 0$ and solve for y.

3. **Symmetry**
   If $f(-x) = f(x)$, then f is an **even function** and the curve is **symmetric about the *y*-axis**.
   If $f(-x) = -f(x)$, then f is an **odd function** and the curve is **symmetric about the origin**.
   When replacing y by $-y$ yields an equivalent equation, the curve is **symmetric about the *x*-axis**.

4. **Asymptote**
   If either $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$, then the line $y = L$ is a **horizontal asymptote** of the curve $y = f(x)$.
   If one of the following is true: $\lim_{x\to a^+} f(x) = \infty$, $\lim_{x\to a^-} f(x) = \infty$, $\lim_{x\to a^+} f(x) = -\infty$ or $\lim_{x\to a^-} f(x) = -\infty$, then the line $x = a$ is a **vertical asymptote** of the curve $y = f(x)$.

5. **Intervals of Increasing or Decreasing**
   Compute $f'(x)$. If $f'(x) > 0$ f is increasing and if $f'(x) < 0$ f is decreasing.

6. **Relative Maximum and Relative Minimum Values**
   Find the critical numbers then use the first derivative test.

7. **Points of Inflection and Concavity**
   Compute $f''(x)$. Inflection point occur where the direction of concavity changes. The curve is concave upward where $f''(x) > 0$ and concave downward where $f''(x) < 0$.

Exercises

1. The graph of $f(x) = xe^{-x^2}$ is symmetric about which of the following

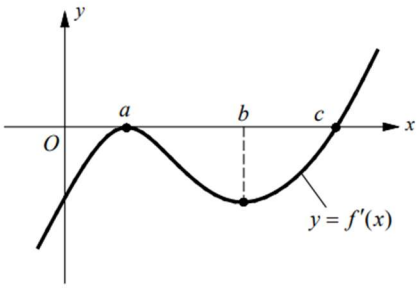
I. The *x*-axis
II. The *y*-axis
III. The origin

(A) I only (B) II only (C) III only (D) II and III only

2. Let f be the function given by $f(x)=\dfrac{-3x^2}{\sqrt{3x^4+1}}$. Which of the following is the equation of horizontal asymptote of the graph of f ?

(A) $y=-3$ (B) $y=-\sqrt{3}$ (C) $y=\sqrt{3}$ (D) $y=3$

3.



The graph of f', the derivative of function f, is shown above. If f is a twice differentiable function, which of the following statements must be true?

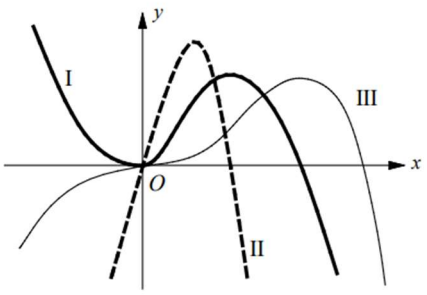
I. $f(c)>f(a)$

II. The graph of f is concave up on the interval $b<x<c$.

III. f has a relative minimum at $x=c$.

(A) I only (B) II only (C) III only (D) II and III only

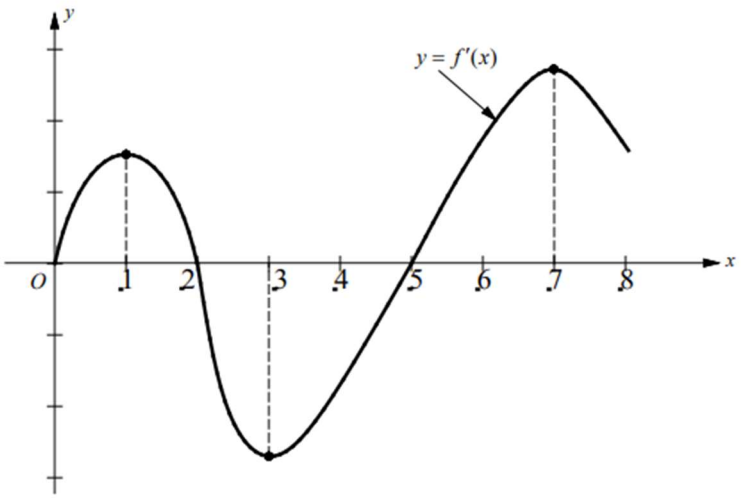
4.



Three graphs labeled I, II, and III are shown above. They are the graphs of f, f', and f''. Which of the following correctly identifies each of the three graphs?

	f	f'	f''
(A)	I	II	III
(B)	II	I	III
(C)	III	I	II
(D)	I	III	II

5.



The figure above shows the graph of f', the derivative of a function f. The domain of f is the set of all real numbers x such that $0 \le x \le 8$.

(a) For what values of x does the graph of f have a horizontal tangent?

(b) On what intervals is f increasing?

(c) On what intervals is f concave upward?

(d) For what values of x does the graph of f have a relative maximum?

(e) Find the x-coordinate of each inflection point on the graph of f.

3.7 Optimization Problems

When you are given a word problem that asks for the maximum or minimum value of a certain quantity, you have to translate the word problem into a mathematical equation for the quantity that is to be maximized or minimized. The differential calculus is a powerful tool for solving these kind of problems.

Guidelines for Solving Optimization Problems

1. Read the problem carefully until you understand it.
2. In most problems it is useful to draw a picture. Label it with the quantities given in the problem.
3. Assign a variable to the unknown quantity and write an equation for the quantity that is to be maximized (or minimized), since this equation will usually involve two or more variables.
4. Use the given information to find relationships between these variables. Use these equations to eliminate all but one variable in the equation.
5. Use the first and second derivatives tests to find the critical points.

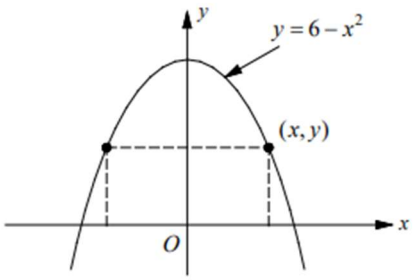
Exercises

1.

The point on the curve $y = 2 - x^2$ nearest to $(3,2)$ is

(A) $(0,2)$ (B) $(\frac{1}{2},\frac{7}{4})$ (C) $(\frac{3}{4},\frac{23}{16})$ (D) $(1,1)$

2.



What is the area of the largest rectangle that has its base on the x-axis and its other two vertices on the parabola $y = 6 - x^2$?

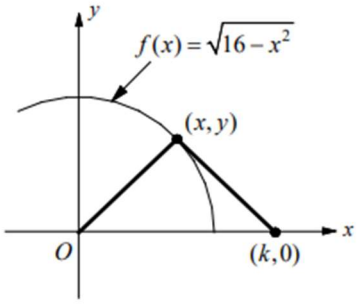
(A) $8\sqrt{2}$ (B) $6\sqrt{2}$ (C) $4\sqrt{3}$ (D) $3\sqrt{2}$

3.

If $y = \frac{1}{\sqrt{x}} - \sqrt{x}$, what is the maximum value of the product of xy ?

(A) $\frac{1}{9}$ (B) $\frac{\sqrt{3}}{9}$ (C) $\frac{2\sqrt{3}}{9}$ (D) $\frac{2}{3}$

4.



Let $f(x) = \sqrt{16 - x^2}$. An isosceles triangle, whose base is the line segment from $(0,0)$ to $(k,0)$, where $k > 0$, has its vertex on the graph of f as shown in the figure above.

(a) Find the area of the triangle in terms of k .

(b) For what values of k does the triangle have a maximum area?

Integration

Definition of an Antiderivative

A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x on I.

Representation Antiderivatives

If F is an antiderivative of f on an interval I, then $F(x)+C$ represents the most general antiderivative of f on I, where C is a constant.

Definition of Indefinite Integral

The set of all antiderivatives of f is the indefinite integral of f with respect to x denoted by

$$\int f(x)dx .$$

Thus $\int f(x)dx = F(x)+C$ means $F'(x) = f(x)$.

Table of Indefinite Integrals

$$\int k\ dx = kx + C \qquad \int k\,f(x)\ dx = k\int f(x)\ dx$$

$$\int \left[f(x) \pm g(x)\right] dx = \int f(x)\ dx \pm \int g(x)\ dx$$

$$\int x^n\ dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1 \qquad \text{Power Rule}$$

$$\int e^x\ dx = e^x + C$$

$$\int \sin x\ dx = -\cos x + C \qquad \int \cos x\ dx = \sin x + C$$

$$\int \sec^2 x\ dx = \tan x + C \qquad \int \csc^2 x\ dx = -\cot x + C$$

$$\int \sec x \tan x\ dx = \sec x + C \qquad \int \csc x \cot x\ dx = -\csc x + C$$

Exercises

1. If $\frac{dy}{dx} = 3x^2 - 1$, and if $y = -1$ when $x = 1$, then $y =$

(A) $x^3 - x + 1$

(B) $x^3 - x - 1$

(C) $-x^3 + x - 1$

(D) $-x^3 + 1$

2. Which of the following is the antiderivative of $f(x) = \tan x$?

(A) $\sec x + \tan x + C$

(B) $\csc x + \cot x + C$

(C) $\ln|\csc x| + C$

(D) $-\ln|\cos x| + C$

3. A curve has a slope of $-x + 2$ at each point (x, y) on the curve. Which of the following is an equat for this curve if it passes through the point $(2,1)$?

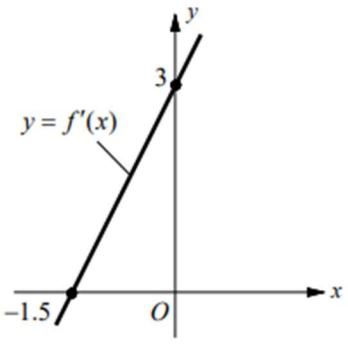
(A) $\frac{1}{2}x^2 - 2x - 4$

(B) $2x^2 + x - 8$

(C) $-\frac{1}{2}x^2 + 2x - 1$

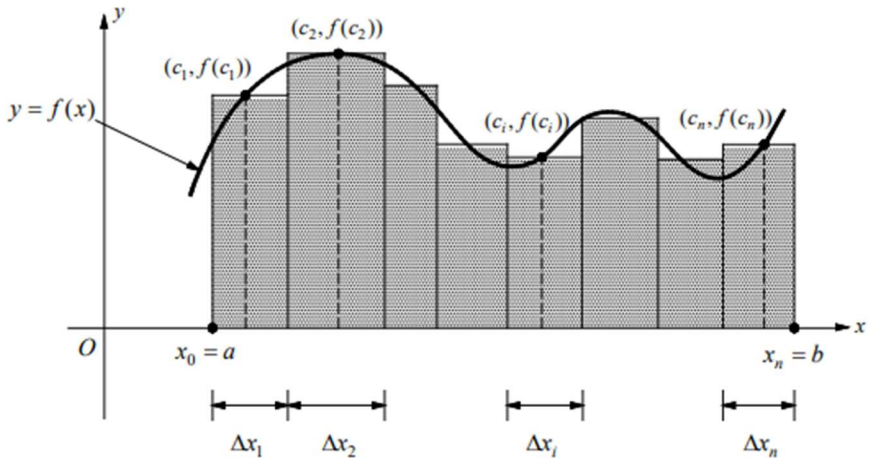
(D) $x^2 - 2x + 1$

4.



The graph of f', the derivative of f, is the line shown in the figure above. If $f(3) = 11$, the

The **area of a region *S*** that lies under the curve $y = f(x)$ from a to b can be approximated by summing the areas of a collection of rectangles.



Definition of Riemann Sum

Let f be a continuous function defined on the closed interval $[a,b]$, and let Δ be a partition of $[a,b]$ given by

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

where Δx_i is the width of the i th interval. If c_i is any point in the i th interval, then the sum

$$\sum_{i=1}^{n} f(c_i)\Delta x_i = f(c_1)\Delta x_1 + f(c_2)\Delta x_2 + \cdots + f(c_i)\Delta x_i + \cdots + f(c_n)\Delta x_n$$

is called a **Riemann sum** for f on the interval $[a,b]$.

If every subinterval is of equal width, the partition is **regular** and $\Delta x = \dfrac{b-a}{n}$.

Then the Riemann sum can be written

$$\sum_{i=1}^{n} f(c_i)\Delta x = \Delta x\left[f(c_1) + f(c_2) + \cdots + f(c_i) + \cdots + f(c_n)\right]$$

where $c_i = a + i(\Delta x)$.

Left, Right, and Midpoint Riemann Sum Approximation

If c_i is the left endpoint of each subinterval then $\sum_{i=1}^{n} f(c_i)\Delta x_i$ is called a **left Riemann sum**.

If c_i is the right endpoint of each subinterval then $\sum_{i=1}^{n} f(c_i)\Delta x_i$ is called a **right Riemann sum**.

If c_i is the midpoint of each subinterval then $\sum_{i=1}^{n} f(c_i)\Delta x_i$ is called a **midpoint Riemann sum**.

Exercises

1. Using a left Riemann sum with three subintervals $[0,1]$, $[1,2]$, and $[2,3]$, what is the approximation of $\int_0^3 (3-x)(x+1)\,dx$?

(A) 7.5 (B) 9 (C) 10 (D) 11.5

2. The expression $\frac{1}{20}\left[\left(\frac{1}{20}\right)^2+\left(\frac{2}{20}\right)^2+\left(\frac{3}{20}\right)^2+\ldots+\left(\frac{20}{20}\right)^2\right]$ is a Riemann sum approximation for

(A) $\frac{1}{20}\int_0^{20} x^2\,dx$

(B) $\frac{1}{20}\int_0^{1} x^2\,dx$

(C) $\int_0^{1} x^2\,dx$

(D) $\int_0^{1} \frac{1}{x^2}\,dx$

3. Using a midpoint Riemann sum with three subintervals $[0,1]$, $[1,2]$, and $[2,3]$, what is the approximation of $\int_0^3 \sqrt{1+x^2}\,dx$?

(A) 5.613 (B) 6.213 (C) 6.812 (D) 7.195

4. The expression $\frac{1}{10}\left[\frac{1}{10}+\frac{2}{10}+\frac{3}{10}+\ldots+\frac{20}{10}\right]$ is a Riemann sum approximation for

(A) $\int_0^2 2\,dx$

(B) $\int_0^2 x\,dx$

(C) $\int_0^2 \frac{x}{10}\,dx$

(D) $\frac{1}{10}\int_0^1 x\,dx$

Definition of a Definite Integral

If f is a continuous function defined for $a \le x \le b$, then the definite integral of f from a to b is

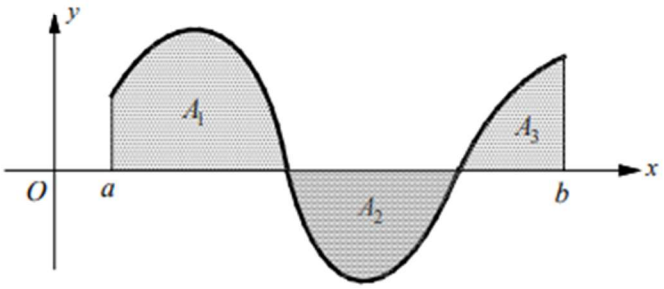
$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\Delta x,$$

where $c_i = a + i\Delta x$ and $\Delta x = (b-a)/n$.

If $y = f(x)$ is continuous and nonnegative over a closed interval $[a,b]$ then the **area** of the region bounded by the graph of f, the x-axis, and the vertical lines $x = a$ and $x = b$ is given by

$$\textbf{Area} = \int_a^b f(x)\,dx.$$

If $f(x)$ takes on both positive and negative values over a closed interval $[a,b]$, then the area of the region bounded by the graph of f and the x-axis is obtained by adding the absolute value of the definite integral over each subinterval where $f(x)$ does not change sign.



The figure above shows how both the total area and the value of definite integral can be interpreted in terms of areas between the graph of $f(x)$ and the x-axis.

The definite integral of $f(x)$ over $[a,b] = \int_a^b f(x)\,dx = A_1 - A_2 + A_3$.

The total area between the curve and the x-axis over $[a,b] = \int_a^b |f(x)|\,dx = A_1 + A_2 + A_3$.

Application of Definite Integral

If the population density of a region at a distance x from a straight road is $D(x)$, then the **total population** of the region between $x = a$ and $x = b$ is given by

$$\textbf{Total Population} = \int_a^b f(x)\cdot D(x)\,dx.$$

If the population density of a circular region at a distance r from the center is $D(r)$, then the total population of the circular region between $r = a$ and $r = b$ is given by

$$\textbf{Total Population} = 2\pi\int_a^b r\,D(r)\,dr.$$

Exercises

1. $\int_0^3 \frac{dx}{\sqrt{1+x}} =$

(A) 2 (B) 2.5 (C) 3 (D) 4

2. The area of the region in the first quadrant enclosed by the graph of $f(x) = 4x - x^3$ and the x-axis is

(A) $\frac{11}{4}$ (B) $\frac{7}{2}$ (C) 4 (D) $\frac{11}{2}$

3. The population density of a circular region is given by $f(r) = 10 - 3\sqrt{r}$ people per square mile, where r is the distance from the center of the city, in miles. Which of the following expressions gives the number of people who live within a 3 mile radius from the center of the city?

(A) $\pi \int_0^3 r^2(10 - 3\sqrt{r})\, dr$

(B) $\pi \int_0^3 (r+3)^2(10 - 3\sqrt{r})\, dr$

(C) $2\pi \int_0^3 (r+3)(10 - 3\sqrt{r})\, dr$

(D) $2\pi \int_0^3 r(10 - 3\sqrt{r})\, dr$

4. Which of the following limits is equal to $\int_1^3 x^3\, dx$?

(A) $\lim_{n\to\infty} \sum_{i=1}^{n} (1 + \frac{i}{n})^3 \frac{1}{n}$

(B) $\lim_{n\to\infty} \sum_{i=1}^{n} (1 + \frac{i}{n})^3 \frac{2}{n}$

(C) $\lim_{n\to\infty} \sum_{i=1}^{n} (1 + \frac{2i}{n})^3 \frac{1}{n}$

(D) $\lim_{n\to\infty} \sum_{i=1}^{n} (1 + \frac{2i}{n})^3 \frac{2}{n}$

5. Which of the following integrals is equal to $\lim_{n\to\infty}\sum_{i=1}^{n}(-1+\frac{3i}{n})^2\frac{3}{n}$?

(A) $\int_{-1}^{2} x^2\,dx$

(B) $\int_{-1}^{0} x^2\,dx$

(C) $\int_{-1}^{2} (-1+x)^2\,dx$

(D) $\int_{-1}^{0} (-1+\frac{x}{3})^2\,dx$

6. The closed interval $[a,b]$ is partitioned into n equal subintervals, each of width Δx, by the numbers $x_0, x_1, \ldots, x_n$ where $0 < a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$. What is $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{1}{\sqrt{x_i}}\Delta x$?

(A) $\frac{1}{\sqrt{b}}-\frac{1}{\sqrt{a}}$

(B) $\frac{(\sqrt{b}-\sqrt{a})}{2}$

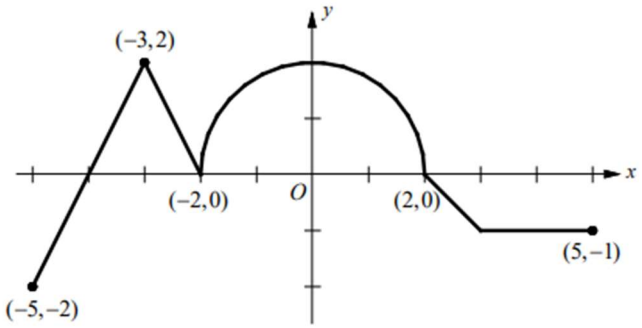
(C) $2(\sqrt{b}-\sqrt{a})$

(D) $\sqrt{b}-\sqrt{a}$

7. If n is a positive integer, then $\lim_{n\to\infty}\frac{1}{n}\left[\left(\frac{1}{n}\right)^2+\left(\frac{2}{n}\right)^2+\cdots+\left(\frac{n}{n}\right)^2\right]$ can be expressed as

(A) $\int_{0}^{1}\frac{1}{x}\,dx$ (B) $\int_{0}^{1}\frac{1}{x^2}\,dx$ (C) $\int_{0}^{1}x^2\,dx$ (D) $\frac{1}{2}\int_{0}^{1}x^2\,dx$

8.



The graph of $y = f(x)$ consists of four line segments and a semicircle as shown in the figure above. Evaluate each definite integral by using geometric formulas.

(a) $\int_{-5}^{-2} f(x)\,dx$ (b) $\int_{-2}^{2} f(x)\,dx$ (c) $\int_{2}^{5} f(x)\,dx$ (d) $\int_{-5}^{5} |f(x)|\,dx$

Properties of Definite Integral

Definition

$\int_a^a f(x)\,dx = 0$ $\qquad$ $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

Constant Multiple

$\int_a^b c\,dx = c(b-a)$ $\qquad$ $\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$

Sum and Difference

$\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$

Additivity

$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$

Integrals of Symmetric Functions

If f is even $f(-x) = f(x)$, then $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$

If f is odd $f(-x) = -f(x)$, then $\int_{-a}^a f(x)\,dx = 0$

Comparison Property

If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x)\,dx \geq 0$

If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$

If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then $m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$

Exercises

1. If $\int_a^b f(x)\,dx = 2a - 5b$, then $\int_a^b [f(x) - 2\,]dx =$

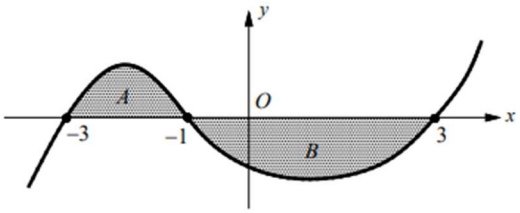
(A) $-7b$ (B) $-3b$ (C) $4a - 7b$ (D) $4a - 3b$

2. If $\int_1^6 f(x)\,dx = \frac{15}{2}$ and $\int_6^4 f(x)\,dx = 5$, then $\int_1^4 f(x)\,dx =$

(A) $\frac{5}{2}$ (B) $\frac{9}{2}$ (C) $\frac{19}{2}$ (D) $\frac{25}{2}$

3. If $\int_{-2}^{6} f(x)\,dx = 10$ and $\int_{2}^{6} f(x)\,dx = 3$, then $\int_{2}^{6} f(4-x)\,dx =$

(A) 3 (B) 6 (C) 7 (D) 10

4. 

The graph of $y = f(x)$ is shown in the figure above. If A and B are positive numbers that represent the areas of the shaded regions, what is the value of $\int_{-3}^{3} f(x)\,dx - 2\int_{-1}^{3} f(x)\,dx$, in terms of A and B?

(A) $-A-B$ (B) $A+B$ (C) $A-2B$ (D) $A-B$

5. Let f and g be a continuous function on the interval $[1,5]$. Given $\int_{1}^{3} f(x)\,dx = -3$, $\int_{1}^{5} f(x)\,dx = 7$, and $\int_{1}^{5} g(x)\,dx = 9$, find the following definite integrals.

(a) $\int_{3}^{5} f(x)\,dx$ (b) $\int_{1}^{3} [f(x)+3]\,dx$ (c) $\int_{5}^{1} 2g(x)\,dx$

(d) $\int_{5}^{5} g(x)\,dx + \int_{5}^{3} f(x)\,dx$ (e) $\int_{-1}^{3} f(x+2)\,dx$

4.5 Trapezoidal Rule

Let f be continuous on $[a,b]$. The **trapezoidal rule** for approximating $\int_a^b f(x)\,dx$ is given by

$$\int_a^b f(x)\,dx = \frac{\Delta x}{2}\left[f(x_0)+2f(x_1)+2f(x_2)+\cdots+2f(x_{n-1})+f(x_n)\right],$$

where $x_0 = a$, $x_1 = a+\Delta x$, $x_2 = a+2\Delta x$, $\ldots$, $x_{n-1} = a+(n-1)\Delta x$, $x_n = b$, and $\Delta x = \dfrac{b-a}{n}$.

Exercises

1. If four equal subdivisions on $[0,2]$ are used, what is the trapezoidal approximation of $\int_0^2 e^x\,dx$?

(A) $\frac{1}{4}\left[1+2\sqrt{e}+2e+2e\sqrt{e}+e^2\right]$

(B) $\frac{1}{2}\left[1+2\sqrt{e}+2e+2e\sqrt{e}+e^2\right]$

(C) $\frac{1}{4}\left[1+\sqrt{e}+e+e\sqrt{e}+e^2\right]$

(D) $\frac{1}{2}\left[1+\sqrt{e}+e+e\sqrt{e}+e^2\right]$

2. If three equal subdivisions on $\left[\frac{\pi}{2},\pi\right]$ are used, what is the trapezoidal approximation of $\int_{\pi/2}^{\pi} \sin x\,dx$?

(A) $\frac{\pi}{12}(\sin\frac{2\pi}{3}+\sin\frac{5\pi}{6}+\sin\pi)$

(B) $\frac{\pi}{12}(\sin\frac{\pi}{2}+\sin\frac{2\pi}{3}+\sin\frac{5\pi}{6}+\sin\pi)$

(C) $\frac{\pi}{12}(\sin\frac{\pi}{2}+2\sin\frac{2\pi}{3}+2\sin\frac{5\pi}{6}+\sin\pi)$

(D) $\frac{\pi}{6}(\sin\frac{\pi}{2}+\sin\frac{2\pi}{3}+\sin\frac{5\pi}{6}+\sin\pi)$

3. Find a trapezoidal approximation of $\int_0^2 \cos(x^2)\,dx$ using four subdivisions of length $\Delta x = 0.5$.

4.6 Fundamental Theorem of Calculus Part 1

If f is continuous on $[a,b]$ then $F(x)=\int_a^x f(t)\,dt$ is continuous on $[a,b]$ and differentiable on (a,b), so

$$F'(x)=\frac{d}{dx}\int_a^x f(t)\,dt=f(x).$$

If $F(x)=\int_a^{u(x)} f(t)\,dt$, then by the chain rule

$$F'(x)=\frac{d}{dx}\int_a^{u(x)} f(t)\,dt=f(u)\cdot\frac{du}{dx}.$$

Exercises

1. $\dfrac{d}{dx}\displaystyle\int_1^{x^2}\sqrt{3+t^2}\,dt=$

(A) $\sqrt{3+x^2}$ (B) $\sqrt{3+x^4}$ (C) $2x\sqrt{3+x^4}$ (D) $2\sqrt{3+x^2}$

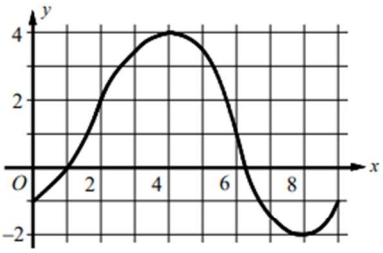
2. For $-\dfrac{\pi}{2}<x<\dfrac{\pi}{2}$, if $F(x)=\displaystyle\int_0^{\sin x}\frac{dt}{\sqrt{1-t^2}}$, then $F'(x)=$

(A) $\dfrac{\sin x}{\sqrt{1-x^2}}$ (B) $\dfrac{\cos x}{\sqrt{1-x^2}}$ (C) 1 (D) $\csc x$

3. If $F(x)=\displaystyle\int_0^{\sqrt{x}}\cos(t^2)\,dt$, then $F'(4)=$

(A) $\cos 2$ (B) $\dfrac{\cos 4}{4}$ (C) $\dfrac{\cos 4}{\sqrt{2}}$ (D) $\sqrt{2}\cos 4$

4.



graph of g

The graph of the function g, shown in the figure above, has horizontal tangents at $x=4$ and $x=8$. If $f(x)=\displaystyle\int_0^{\sqrt{x}} g(t)\,dt$, what is the value of $f'(4)$?

(A) 0 (B) $\dfrac{1}{2}$ (C) $\dfrac{3}{4}$ (D) $\dfrac{3}{2}$

4.7 Fundamental Theorem of Calculus Part 2

If f is continuous on $[a,b]$ and F is an antiderivative of f on $[a,b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Exercises

1. If f is the antiderivative of $\dfrac{\sqrt{x}}{1+x^3}$ such that $f(1) = 2$, then $f(3) =$

(A) 1.845 (B) 2.397 (C) 2.906 (D) 3.234

2. If $f(x) = \sqrt{x^4 - 3x + 4}$ and g is the antiderivative of f, such that $g(3) = 7$, then $g(0) =$

(A) −2.966 (B) −1.472 (C) −0.745 (D) 1.086

3. If f is a continuous function and $F'(x) = f(x)$ for all real numbers x, then $\int_2^{10} f(\frac{1}{2}x)\,dx =$

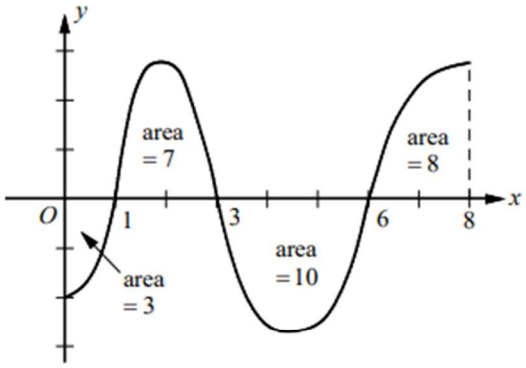
(A) $\frac{1}{2}[F(5) - F(1)]$

(B) $\frac{1}{2}[F(10) - F(2)]$

(C) $2[F(5) - F(1)]$

(D) $2[F(10) - F(2)]$

4.



The figure above shows the graph of f', the derivative of a differentiable function f, on the clc interval $0 \le x \le 8$. The areas of the regions between the graph of f' and the x-axis are labeled i figure. Given $f(6) = 9$, find each of the following.

(a) $f(0)$ (b) $f(1)$ (c) $f(3)$ (d) $f(8)$

Integration by Substitution

The Substitution Rule

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

Evaluating the integral $\int f(g(x))g'(x)\,dx$, when f and g' are continuous functions

1. Substitute $u = g(x)$ and $du = g'(x)\,dx$ to obtain the integral $\int f(u)\,du$.
2. Integrate with respect to u.
3. Replace u by $g(x)$ in the result.

The Substitution Rule for Definite Integrals

If g' is continuous on $[a,b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))\cdot g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

Exercises

1. $\int \sqrt{x}\sin(x^{3/2})\,dx =$

(A) $\frac{2}{3}\cos(x^{3/2})+C$

(B) $\frac{2}{3}\sqrt{x}\cos(x^{3/2})+C$

(C) $-\frac{2}{3}x^{3/2}\cos(x^{3/2})+C$

(D) $-\frac{2}{3}\cos(x^{3/2})+C$

2. If $\int_{-1}^{3} f(x+k)\,dx = 8$, where k is a constant, then $\int_{k-1}^{k+3} f(x)\,dx =$

(A) $8-k$ (B) $8+k$ (C) 8 (D) $k-8$

3. If the substitution $u = 1+\sqrt{x}$ is made, $\int \frac{(1+\sqrt{x})^{3/2}}{\sqrt{x}}\,dx =$

(A) $\frac{1}{2}\int u^{3/2}\,du$ (B) $2\int u^{3/2}\,du$ (C) $\frac{1}{2}\int \sqrt{u}\,du$ (D) $2\int \sqrt{u}\,du$

4. If the substitution $u = \ln x$ is made, $\int_e^{e^2} \frac{1-(\ln x)^2}{x}\,dx =$

(A) $\int_e^{e^2} (\frac{1}{u}-u^2)\,du$

(B) $\int_e^{e^2} (\frac{1}{u}-u)\,du$

(C) $\int_1^2 (1-u^2)\,du$

(D) $\int_1^2 (1-u)\,du$

Integration of Exponential and Logarithmic Function

Definition of the Natural Logarithmic Function

The **natural logarithmic function** is defined by

$$\ln x = \int_1^x \frac{1}{t}\,dt, \qquad x > 0$$

If u is a differentiable function such that $u \neq 0$,

$$\int \frac{1}{u}\,du = \ln|u| + C.$$

Whether u is positive or negative, the integral of $(1/u)\,du$ is $\ln|u| + C$.

Integration of Exponential Function

$$\int e^u\,du = e^u + C$$

Exercises

1. $\int_1^3 \frac{x+3}{x^2+6x}\,dx =$

(A) $\ln\frac{3}{2}$ (B) $\frac{\ln 27 - \ln 7}{2}$ (C) $\ln 3$ (D) $\frac{\ln 20 - \ln 5}{2}$

2. $\int_0^{\pi/2} \cos x\, e^{\sin x}\,dx =$

(A) $-e$ (B) $1-e$ (C) $\frac{e}{2}$ (D) $e-1$

3. Let $F(x)$ be an antiderivetive of $\ln(\sin^2 x)+3$. If $F(1)=2$, then $F(3)=$

(A) 6.595 (B) 7.635 (C) 10.036 (D) 12.446

4. $\int_1^e \frac{\cos(\ln x)}{x}\,dx =$

(A) $\frac{1}{\sin 1}$ (B) $\frac{1}{\cos 1}$ (C) $\sin(e)$ (D) $\sin 1$

Applications of Integration

Area of a Region Between Two Curves

If f and g are continuous on $[a,b]$ and $g(x) \le f(x)$ for all x on $[a,b]$, then the area of the region bounded by the graphs of f and g and the vertical lines $x = a$ and $x = b$ is

$$A = \int_a^b [f(x) - g(x)]\, dx .$$

In general, to determine the area between two curves, we can use

$$A = \int_{x_1}^{x_2} [\textbf{(top curve)} - \textbf{(bottom curve)}]\, dx ,$$

if the curves are defined by functions of x, and

$$A = \int_{y_1}^{y_2} [\textbf{(right curve)} - \textbf{(left curve)}]\, dy ,$$

if the curves are defined by functions of y.

Exercises

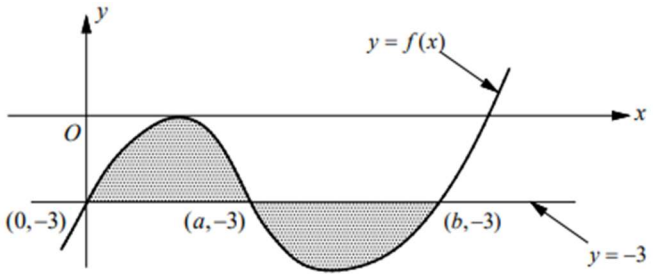
1. What is the area of the region enclosed by the graphs of $f(x) = x + 2$ and $g(x) = x^3 - 4x^2 + 6$?

(A) $\frac{193}{12}$ (B) $\frac{218}{12}$ (C) $\frac{253}{12}$ (D) $\frac{305}{12}$

2. What is the area of the region in the first quadrant, bounded by the curve $y = \sqrt[3]{x}$ and $y = x$?

(A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

3.



The curve $y = f(x)$ and the line $y = -3$, shown in the figure above, intersect at the points $(0,-3)$, $(a,-3)$, and $(b,-3)$. The sum of area of the shaded region enclosed by the curve and the line is given by

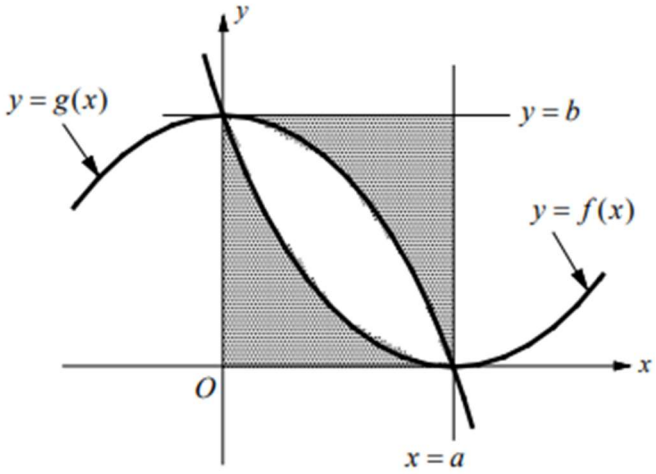
(A) $\int_0^a [3 - f(x)]\,dx + \int_a^b [-3 + f(x)]\,dx$

(B) $\int_0^a [-3 + f(x)]\,dx + \int_a^b [3 - f(x)]\,dx$

(C) $\int_0^a [f(x) + 3]\,dx + \int_a^b [-3 - f(x)]\,dx$

(D) $\int_0^a [f(x) - 3]\,dx + \int_a^b [3 - f(x)]\,dx$

4.



Which of the following is the area of the shaded region in the figure above?

(A) $\int_0^a [g(x) - f(x)]\,dx$

(B) $\int_0^a [b + g(x) - f(x)]\,dx$

(C) $\int_0^a [b - g(x) - f(x)]\,dx$

(D) $\int_0^a [b - g(x) + f(x)]\,dx$

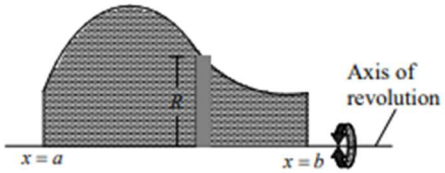
Volumes by Disk, Washer, and Shell

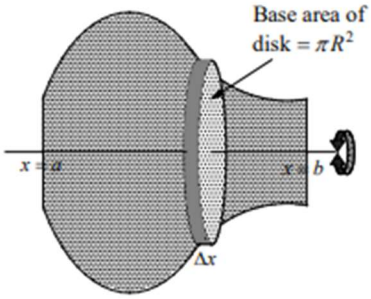
The solid generated by rotating a plane region about an axis is called a **solid of revolution**. The simplest such solid is a right circular cylinder or **disk**.

Volume of disk = (base area of disk)(width of disk)

$\Delta V = A(x)\,\Delta x$.

So the definition of volume gives us $V = \int_a^b A(x)\,dx$.





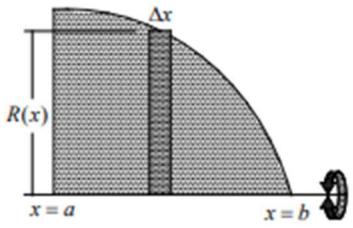
The Disk Method

Horizontal Axis of Revolution

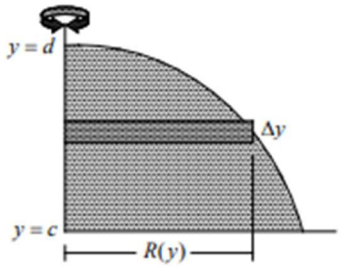
$$V = \pi \int_a^b [R(x)]^2\,dx$$

Vertical Axis of Revolution

$$V = \pi \int_c^d [R(y)]^2\,dy$$



Horizontal axis of revolution



Vertical axis of revolution

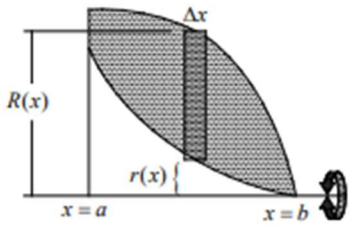
The Washer Method

Horizontal Axis of Revolution

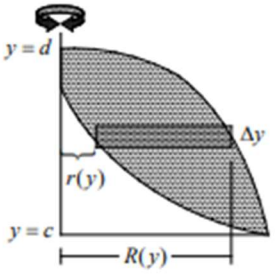
$$V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2)\,dx$$

Vertical Axis of Revolution

$$V = \pi \int_c^d ([R(y)]^2 - [r(y)]^2)\,dy$$

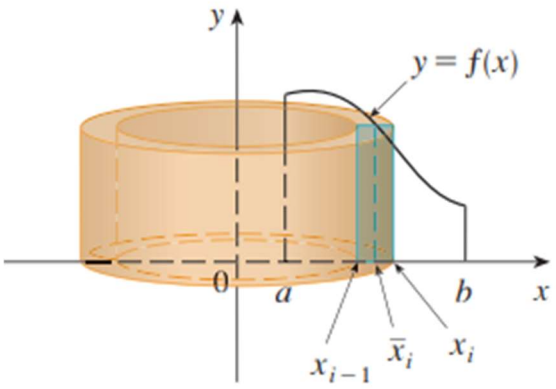


Horizontal axis of revolution



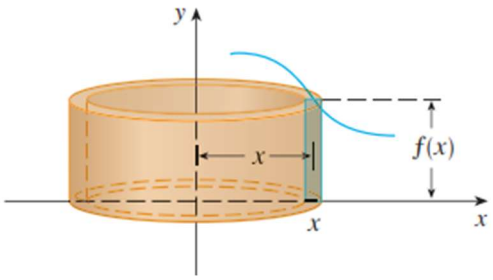
Vertical axis of revolution

Shell Method

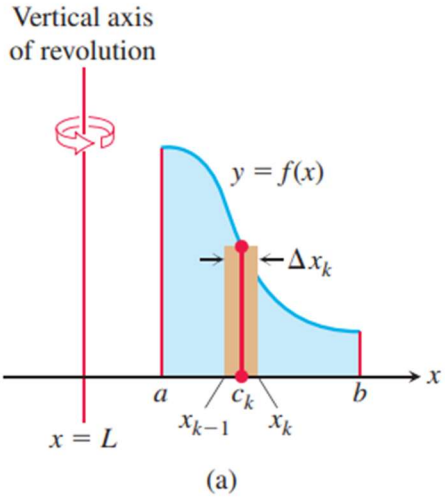


2 The volume of the solid in Figure 3, obtained by rotating about the y-axis the region under the curve $y = f(x)$ from a to b, is

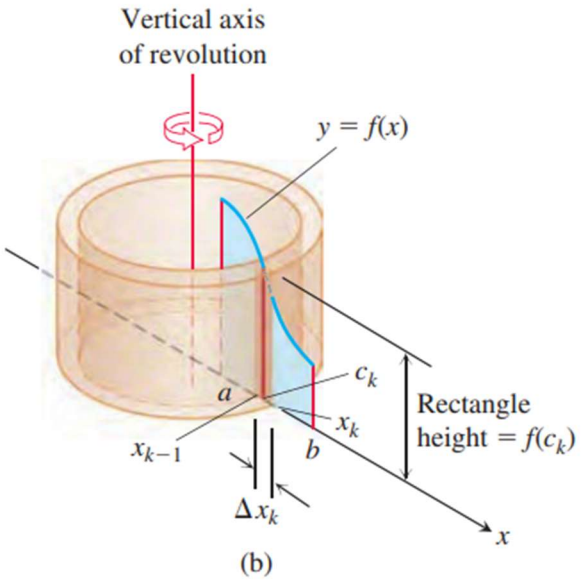
$$V = \int_a^b 2\pi x f(x)\,dx \qquad \text{where } 0 \leqslant a < b$$







(a)

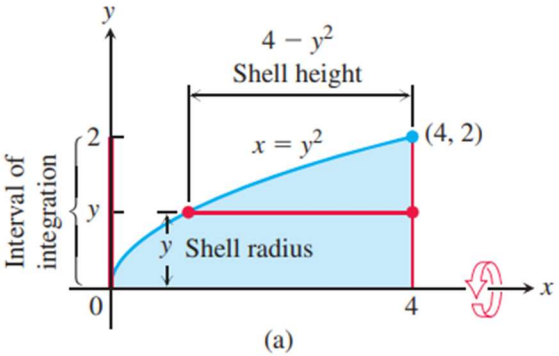


(b)

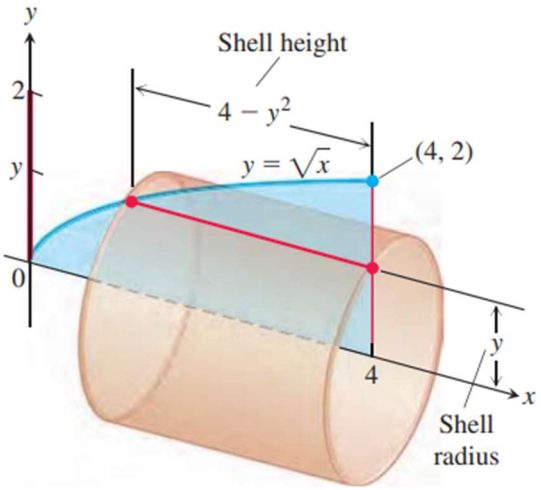
Shell Formula for Revolution About a Vertical Line

The volume of the solid generated by revolving the region between the x-axis and the graph of a continuous function $y = f(x) \geq 0$, $L \leq a \leq x \leq b$, about a vertical line $x = L$ is

$$V = \int_a^b 2\pi \left(\begin{array}{c} \text{shell} \\ \text{radius} \end{array} \right) \left(\begin{array}{c} \text{shell} \\ \text{height} \end{array} \right) dx.$$



(a)



(b)

Exercises

1. The region in the first quadrant bounded by the graph of $y = \sec x$, $x = \frac{\pi}{3}$, and the coordinate axes is rotated about the x-axis. What is the volume of the solid generated?

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\sqrt{3}\pi$ (D) 3π

2. The region enclosed by the graphs of $y = e^{(x/2)}$ and $y = (x-1)^2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. What is the volume of the solid generated?

(A) $\frac{11}{4}\pi$ (B) $2(e-1)\pi$ (C) $(e-\frac{3}{2})\pi$ (D) $(e-\frac{6}{5})\pi$

3. Let R be the region between the graphs of $y = 1+\sin(\pi x)$ and $y = x^3$ from $x = 0$ to $x = 1$. The volume of the solid obtained by revolving R about the x-axis is given by

(A) $\pi\int_0^1 \left[1+\sin(\pi x)-x^3\right] dx$

(B) $\pi\int_0^1 \left[(1+\sin(\pi x))^2 - x^6\right] dx$

(C) $\pi\int_0^1 \left[1+\sin(\pi x)-x^3\right]^2 dx$

(D) $2\pi\int_0^1 \left[1+\sin(\pi x)-x^3\right] dx$

4. The region R is enclosed by the graph of $y = 3x - x^2$ and the line $y = x$. If the region R is rotated about the line $y = -1$, the volume of the solid that is generated is represented by which of the following integrals?

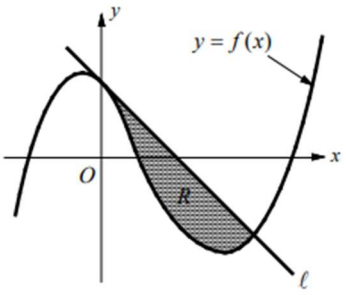
(A) $\pi\int_0^2 \left[3x - x^2 - x + 1\right]^2 dx$

(B) $\pi\int_0^2 \left[\left(3x - x^2 + 1\right)^2 - (x+1)^2\right] dx$

(C) $\pi\int_0^2 \left[\left(3x - x^2 + 1\right) - (x+1)\right]^2 dx$

(D) $\pi\int_0^2 \left[\left(3x - x^2 - 1\right)^2 - (x-1)^2\right] dx$

5.



Let f be the function given by $f(x) = x^3 - 2x^2 - x + \cos x$. Let R be the shaded region bounded by the graph of f and the line ℓ, which is the line tangent to the graph of f at $x = 0$, as shown above.

(a) Find the equation of the line ℓ.

(b) Find the area of R.

(c) Set up, but do not evaluate, an integral expression for the volume of the solid generated when R is revolved about the line $y = 2$.

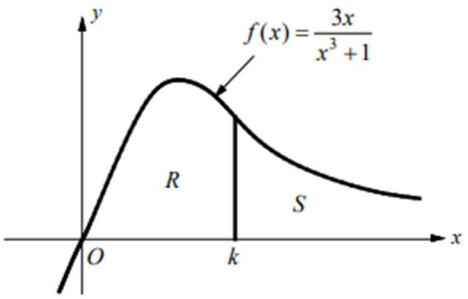
6. Let R be the region between the graphs of $y = e^x$, $y = 2$ and $x = -1$.

(a) Find the area of R.

(b) Find the volume of the solid generated when R is revolved about the line $x = -1$.

(c) Find the volume of the solid generated when R is revolved about the line $y = -1$.

7.



Let f be the function given by $f(x)=\dfrac{3x}{x^3+1}$. Let R be the region bounded by the graph of f, the x-axis, and the vertical line $x=k$, where $k>0$. BC

(a) Find the volume of the solid generated when R is revolved about the x-axis in terms of k.

(b) Let S be the unbounded region in the first quadrant to the right of the vertical line $x=k$ and below the graph of f, as shown in the figure above. Find the value of k such that the volume of the solid generated when S is revolved about the x-axis is equal to the volume of the solid found in part (a).

8. **9–14** Use the method of cylindrical shells to find the volume of the solid obtained by rotating the region bounded by the given curves about the x-axis. Sketch the region and a typical shell.

9. $x = 1 + y^2, \quad x = 0, \quad y = 1, \quad y = 2$

10. $x = \sqrt{y}, \quad x = 0, \quad y = 1$

11. $y = x^3, \quad y = 8, \quad x = 0$

12. $x = 4y^2 - y^3, \quad x = 0$

13. $x = 1 + (y - 2)^2, \quad x = 2$

14. $x + y = 3, \quad x = 4 - (y - 1)^2$

Volumes of Solids with Known Cross Sections

The volume of a solid of known cross-sectional area $A(x)$ from $x = a$ to $x = b$ is the integral of A from a to b.

1. For cross sections of area $A(x)$ taken perpendicular to the x-axis, the volume is

$$V = \int_a^b A(x)\,dx.$$

2. For cross sections of area $A(y)$ taken perpendicular to the y-axis, the volume is

$$V = \int_c^d A(y)\,dy.$$

Exercises

1. The base of a solid is the region enclosed by the graph of $y = e^x$, the coordinate axes, and the line $x = 1$. If the cross sections of the solid perpendicular to the x-axis are squares, what is the volume of the solid?

(A) $\frac{e^2}{4}$ (B) $\frac{e^2-1}{2}$ (C) $\frac{e^2+1}{2}$ (D) $e^2 - \frac{1}{2}$

2. The base of a solid is the region enclosed by the graph of $y = \sqrt{x}$, the x-axis, and the line $x = 2$. If each cross section perpendicular to the x-axis is an equilateral triangle , what is the volume of the solid?

(A) $\frac{\sqrt{3}}{8}$ (B) $\frac{\sqrt{3}}{6}$ (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{\sqrt{3}}{2}$

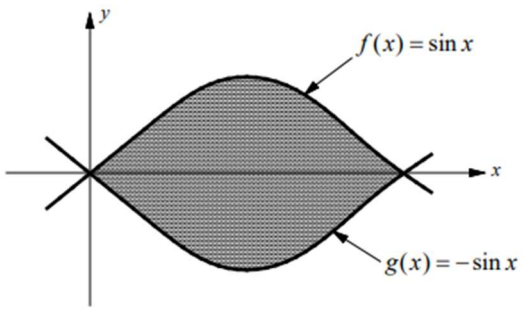
3. The base of a solid S is the semicircular region enclosed by the graph of $y = \sqrt{9-x^2}$ and the x-axis. If the cross sections of S perpendicular to the x-axis are semicircles, what is the volume of the solid?

(A) $\frac{20\pi}{3}$ (B) 6π (C) $\frac{9\pi}{2}$ (D) $\frac{7\pi}{2}$

4. Let R be the region enclosed by the graph of $y = 3\sqrt{x} - x$ and the x-axis. The region R models the surface of a small pond. At all points in R at a distance x from the y-axis, the depth of the water is given by $g(x) = \frac{1}{\sqrt{x}}$. What is the volume of the water in the pond?

(A) $2\sqrt{3}$ (B) 6 (C) $4\sqrt{3}$ (D) 9

5.



Let $f(x) = \sin x$ and $g(x) = -\sin x$ for $0 \le x \le \pi$. The graphs of f and g are shown in the figure above.

(a) Find the area of the shaded region enclosed by the graphs of f and g.

(b) Find the volume of the solid generated when the shaded region enclosed by the graphs of f and g is revolved about the horizontal line $y = 3$.

(c) Let h be the function given by $h(x) = k \sin x$ for $0 \le x \le \pi$. For each $k > 0$, the region (not shown) enclosed by the graphs of h and g is the base of a solid with square cross sections perpendicular to the x-axis. If the volume of the solid is equal to 8π, what is the value of k?

Average Value of a Function

Definition of the Average Value of a Function

If f is integrable on $[a, b]$, then the **average value** of f on the interval is

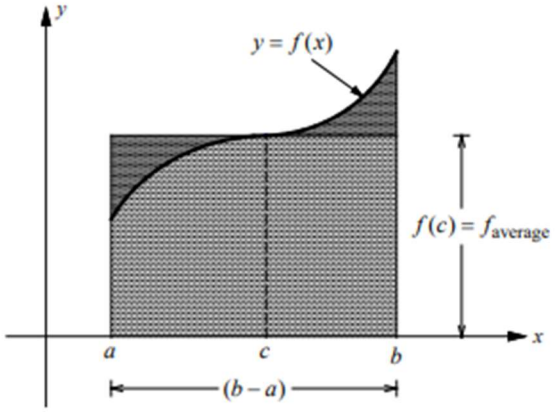
$$\frac{1}{b-a}\int_a^b f(x)\,dx .$$

The Mean Value Theorem for Definite Integrals

If f is continuous on $[a, b]$, then there exists a number c in $[a,b]$, such that

$$\int_a^b f(x)\,dx = f(c)(b-a) .$$

In the figure when $f \geq 0$, the area of the rectangle, which is $f(c)(b-a)$, is equal to the area of the region bounded by the graph of f, the x-axis, and the vertical lines $x = a$ and $x = b$.

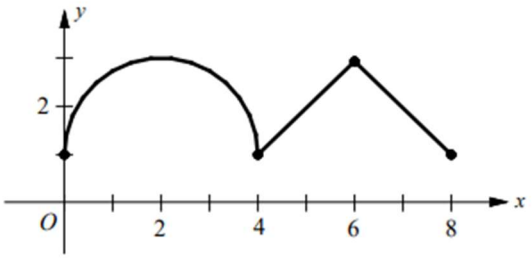


Exercises

1. What is the average value of $f(x) = \sqrt{x}(4-x)$ on the closed interval $[0,4]$?

(A) $\frac{7}{3}$ (B) $\frac{21}{5}$ (C) $\frac{32}{15}$ (D) $\frac{35}{4}$

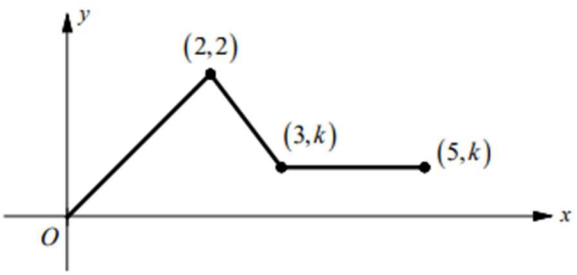
2.



The graph of $y = f(x)$ consists of a semicircle and two line segments. What is the average value of f on the interval $[0,8]$?

(A) $\frac{\pi+2}{4}$ (B) $\frac{\pi+3}{4}$ (C) $\pi+1$ (D) $\frac{\pi+6}{4}$

3.



The graph of $y = f(x)$ consists of three line segments as shown above. If the average value of f on the interval $[0,5]$ is 1 what is the value of k?

(A) $\frac{3}{5}$ (B) $\frac{7}{10}$ (C) $\frac{4}{5}$ (D) $\frac{9}{10}$

Length of a Curve (Distance Traveled Along a Curve)

If f' is continuous on the closed interval $[a,b]$, then the length of the curve $y = f(x)$ from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1+\left[\frac{dy}{dx}\right]^2}\,dx = \int_a^b \sqrt{1+[f'(x)]^2}\,dx$$

If g' is continuous on the closed interval $[c,d]$, then the length of the curve $x = g(y)$ from $y = c$ to $y = d$ is

$$L = \int_c^d \sqrt{1+\left[\frac{dx}{dy}\right]^2}\,dy = \int_c^d \sqrt{1+[g'(y)]^2}\,dy$$

Exercises

1. What is the length of the curve of $y = \frac{1}{3}(x^2+2)^{3/2}$ from $x = 1$ to $x = 2$?

(A) $\frac{8}{3}$ (B) $\frac{10}{3}$ (C) 4 (D) $\frac{14}{3}$

2. Which of the following integrals gives the length of the graph of $y = \ln(\sin x)$ between $x = \frac{\pi}{3}$ to $x = \frac{2\pi}{3}$?

(A) $\int_{\pi/3}^{2\pi/3} \csc^2 x \, dx$

(B) $\int_{\pi/3}^{2\pi/3} \sqrt{1+\cot x} \, dx$

(C) $\int_{\pi/3}^{2\pi/3} \csc x \, dx$

(D) $\int_{\pi/3}^{2\pi/3} \sqrt{1+\csc^2 x} \, dx$

3. Which of the following integrals gives the length of the graph of $y = \frac{1}{3}x^{3/2} - x^{1/2}$ between $x = 1$ to $x = 4$?

(A) $\frac{1}{2}\int_1^4 (\sqrt{x} + \frac{1}{\sqrt{x}}) \, dx$

(B) $\frac{1}{2}\int_1^4 (\sqrt{x} - \frac{1}{\sqrt{x}}) \, dx$

(C) $\frac{1}{2}\int_1^4 (1+\sqrt{x} + \frac{1}{\sqrt{x}}) \, dx$

(D) $\frac{1}{2}\int_1^4 (1+\sqrt{x} - \frac{1}{\sqrt{x}}) \, dx$

Techniques of Integration

Procedures for Fitting Integrands to Basic Rules

Procedure	Example
1. Separating numerator	$\dfrac{1-2x}{1+x^2}=\dfrac{1}{1+x^2}-\dfrac{2x}{1+x^2}$
2. Adding and subtracting terms in numerator	$\dfrac{1}{1-e^x}=\dfrac{1-e^x+e^x}{1-e^x}=\dfrac{1-e^x}{1-e^x}+\dfrac{e^x}{1-e^x}$
3. Dividing improper fractions	$\dfrac{x^3-3x}{x^2-1}=x-\dfrac{2x}{x^2-1}$
4. Completing the square	$\dfrac{1}{\sqrt{4x-x^2}}=\dfrac{1}{\sqrt{4-(x-2)^2}}$

Differentiation Rules

1. $\dfrac{d}{dx}(x^n)=nx^{n-1}$

2. $\dfrac{d}{dx}\left[f(x)g(x)\right]=f(x)g'(x)+g(x)f'(x)$

3. $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right]=\dfrac{g(x)f'(x)-f(x)g'(x)}{\left[g(x)\right]^2}$

4. $\dfrac{d}{dx}f\left[g(x)\right]=f'\left[g(x)\right]g'(x)$

5. $\dfrac{d}{dx}(e^x)=e^x$ 6. $\dfrac{d}{dx}(a^x)=(\ln a)a^x$

7. $\dfrac{d}{dx}\ln|x|=\dfrac{1}{x}$ 8. $\dfrac{d}{dx}\log_a x=\dfrac{1}{\ln a}\cdot\dfrac{1}{x}$

9. $\dfrac{d}{dx}(\sin x)=\cos x$ 10. $\dfrac{d}{dx}(\csc x)=-\csc x\cot x$

Basic Integration Rules

1. $\int du=u+C$

2. $\int u^n\,du=\dfrac{u^{n+1}}{n+1}+C,\quad n\neq -1$

3. $\int \dfrac{du}{u}=\ln|u|+C$

4. $\int e^u\,du=e^u+C$

5. $\int a^u\,du=\dfrac{1}{\ln a}a^u+C$

6. $\int \sin u\,du=-\cos u+C$

7. $\int \cos u\,du=\sin u+C$

8. $\int \sec^2 u\,du=\tan u+C$

13. $\frac{d}{dx}(\tan x) = \sec^2 x$ 14. $\frac{d}{dx}(\cot x) = -\csc^2 x$

15. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

16. $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$

17. $\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$

18. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$

19. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

20. $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$

10. $\int \sec u \tan u \, du = \sec u + C$

11. $\int \csc u \cot u \, du = -\csc u + C$

12. $\int \tan u \, du = -\ln|\cos u| + C$

13. $\int \cot u \, du = \ln|\sin u| + C$

14. $\int \sec u \, du = \ln|\sec u + \tan u| + C$

15. $\int \csc u \, du = -\ln|\csc u + \cot u| + C$

16. $\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\frac{u}{a} + C$

17. $\int \frac{du}{a^2+u^2} = \frac{1}{a}\tan^{-1}\frac{u}{a} + C$

18. $\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{u}{a} + C$

Exercises

1. $\int \frac{1+\sin x}{\cos^2 x}\, dx =$

(A) $\tan x - \sec x \tan x + C$

(B) $\tan x + \sec x + C$

(C) $\tan x + \sec^2 x + C$

(D) $\ln(1+\cos^2 x) + C$

2. $\int \frac{e^{2x}}{1+e^x}\, dx =$

(A) $e^{2x} + \ln(1+e^x) + C$

(B) $e^{2x} - \ln(1+e^x) + C$

(C) $2e^{2x} - \ln(1+e^x) + C$

(D) $e^x - \ln(1+e^x) + C$

3. $\int 2\tan x \ \ln(\cos x)\, dx =$

(A) $\cos x \ [\ln(\cos x)] + C$

(B) $\sin x [\ln(\cos x)] + C$

(C) $-[\ln(\cos x)]^2 + C$

(D) $[\ln(\sin x)]^2 + C$

4. $\int_2^3 \frac{1}{x^2 - 4x + 5}\, dx =$

(A) $\frac{\pi}{4}$ (B) $1 - \frac{\pi}{4}$ (C) $1 + \frac{\pi}{6}$ (D) $1 + \frac{\pi}{4}$

5. $\int \frac{2x}{x^2 + 2x + 1}\, dx =$

(A) $-\operatorname{arc\,cot} x - \frac{1}{x+1} + C$

(B) $\arctan x + \frac{1}{x+1} + C$

(C) $2\ln|x+1| - \frac{2}{(x+1)^2} + C$

(D) $2\ln|x+1| + \frac{2}{x+1} + C$

6. The region bounded by $y = \frac{\sin x}{\sqrt{\cos x}}$, $x = 0$, $x = \frac{\pi}{4}$, and the x-axis is revolved around the x-axis. What is the volume of the resulting solid?

Trigonometric Integrals

Trigonometric Identities

$\sin^2 x + \cos^2 x = 1$ $\quad \tan^2 x + 1 = \sec^2 x$ $\quad \cot^2 x + 1 = \csc^2 x$

$\sin^2 x = \dfrac{1 - \cos 2x}{2}$ $\quad \cos^2 x = \dfrac{1 + \cos 2x}{2}$

$\sin 2x = 2 \sin x \cos x$

Exercises

1. $\displaystyle\int \cos^3 x \sqrt{\sin x}\, dx =$

(A) $\dfrac{1}{3}(\cos x)^3 - \dfrac{2}{5}(\cos x)^{5/2} + C$

(B) $\dfrac{2}{3}(\cos x)^{3/2} - \dfrac{2}{7}(\sin x)^{7/2} + C$

(C) $\dfrac{2}{3}(\sin x)^{3/2} - \dfrac{2}{7}(\sin x)^{7/2} + C$

(D) $\dfrac{2}{3}(\sin x)^{3/2} - \dfrac{2}{5}(\cos x)^{5/2} + C$

2. $\displaystyle\int_0^{\pi} 4\sin^4 \theta\, d\theta =$

(A) π (B) $\dfrac{3\pi}{2}$ (C) 2π (D) $\dfrac{5\pi}{2}$

3. $\displaystyle\int_0^{\pi/4} 4\tan^2 \theta\, d\theta =$

(A) $2 - \pi$ (B) $2 - \dfrac{\pi}{2}$ (C) $2 + \dfrac{\pi}{2}$ (D) $4 - \pi$

4. Find the area bounded by the curves $y = \sin x$ and $y = \sin^3 x$ between $x = 0$ and $x = \dfrac{\pi}{2}$.

6.3 Trigonometric Substitutions

Trigonometric Substitution

1. For integrals involving $\sqrt{a^2-u^2}$, let $u=a\sin\theta$. Then

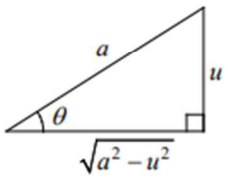
$$\sqrt{a^2-u^2}=\sqrt{a^2-a^2\sin^2\theta}=\sqrt{a^2(1-\sin^2\theta)}=\sqrt{a^2\cos^2\theta}=a\cos\theta$$

2. For integrals involving $\sqrt{a^2+u^2}$, let $u=a\tan\theta$. Then

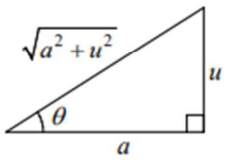
$$\sqrt{a^2+u^2}=\sqrt{a^2+a^2\tan^2\theta}=\sqrt{a^2(1+\tan^2\theta)}=\sqrt{a^2\sec^2\theta}=a\sec\theta$$

3. For integrals involving $\sqrt{u^2-a^2}$, let $u=a\sec\theta$. Then

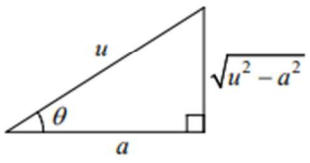
$$\sqrt{u^2-a^2}=\sqrt{a^2\sec^2\theta-a^2+}=\sqrt{a^2(\sec^2\theta-1)}=\sqrt{a^2\tan^2\theta}=a\tan\theta$$



$u=a\sin\theta$



$u=a\tan\theta$



$u=a\sec\theta$

Note: $\operatorname{arcsec}x=\arccos\frac{1}{x}$ $\qquad$ $\operatorname{arccsc}x=\arcsin\frac{1}{x}$

Exercises

1. If the substitution $x=2\tan\theta$ is made in $\int\frac{x^3}{\sqrt{x^2+4}}\,dx$, where $-\frac{\pi}{2}<\theta<\frac{\pi}{2}$, the resulting integral is

(A) $4\int\tan^2\theta\sec\theta\,d\theta$

(B) $4\int\tan^2\theta\sec^2\theta\,d\theta$

(C) $8\int\tan^3\theta\,d\theta$

(D) $8\int\tan^3\theta\sec\theta\,d\theta$

2. $\int_{\sqrt{2}}^{2}\frac{1}{x\sqrt{x^2-1}}\,dx=$

(A) $\frac{\pi}{18}$ $\qquad$ (B) $\frac{\pi}{12}$ $\qquad$ (C) $\frac{\pi}{6}$ $\qquad$ (D) $\frac{\pi}{4}$

3. If the substitution $x = \sec\theta$ is made in $\int \frac{\sqrt{x^2-1}}{x^4}\,dx$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, the resulting integral is

(A) $\int \sec^2\theta\tan\theta\,d\theta + C$

(B) $\int \sec\theta\tan^2\theta\,d\theta + C$

(C) $\int \sin\theta\cos^2\theta\,d\theta + C$

(D) $\int \sin^2\theta\cos\theta\,d\theta + C$

4. The average value of $f(x) = \frac{4}{\sqrt{9+x^2}}$ on the interval $[0,4]$ is

(A) $\ln 2$ (B) $\ln(\sqrt{2}-1)$ (C) $\ln 3$ (D) $\ln(\sqrt{2}+1)$

L'Hospital's Rule

Suppose f and g are differentiable and $g'(x) \neq 0$ near $x = c$ (except possibly at c).

If the limit of $\frac{f(x)}{g(x)}$ as x approaches c produces the indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x\to c}\frac{\boldsymbol{f(x)}}{\boldsymbol{g(x)}} = \lim_{x\to c}\frac{\boldsymbol{f'(x)}}{\boldsymbol{g'(x)}}$$

provided the limit on the right side exists.

L'Hospital's Rule can be applied only to quotients leading to indeterminate forms such as

$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, 1^∞, ∞^0, 0^0, and $\infty - \infty$.

L'Hospital's Rule does not apply when either the numerator or denominator has a finite nonzero limit.

6.5 Integration by Partial Fractions BC

A method for rewriting a rational function into the sum of simpler rational functions is called the **method of partial fractions**. For instance, the rational function $\frac{5x+1}{x^2+x-2}$ can be written as $\frac{5x+1}{x^2+x-2}=\frac{2}{x-1}+\frac{3}{x+2}$.

We call the fractions $\frac{2}{x-1}$ and $\frac{3}{x+2}$ **partial fractions** because their denominators are only part of the original denominator. To integrate the rational function $\frac{5x+1}{x^2+x-2}$, we simply sum the integrals of the partial fractions.

$$\int \frac{5x+1}{x^2+x-2}\,dx=\int \frac{2}{x-1}\,dx+\int \frac{3}{x+2}\,dx=2\ln|x-1|+3\ln|x+2|+C$$

Exercises

1. $\int \frac{dx}{x^2+x-6} =$

(A) $\frac{1}{5}\ln\left|\frac{x-1}{x+6}\right|+C$

(B) $\frac{1}{5}\ln\left|\frac{x+3}{x-2}\right|+C$

(C) $\frac{1}{5}\ln\left|\frac{x-2}{x+3}\right|+C$

(D) $\frac{1}{5}\ln\left|(x-2)(x+3)\right|+C$

2. $\int_4^7 \frac{5}{(x-2)(2x+1)}\,dx =$

(A) $\ln\frac{9}{10}$ (B) $\ln\frac{10}{9}$ (C) $\ln\frac{3}{2}$ (D) $\ln\frac{9}{4}$

3. $\int \frac{x}{x^2+5x+6}\,dx =$

(A) $-2\ln|x+2|+3\ln|(x+3)|+C$

(B) $2\ln|x+2|+3\ln|(x+3)|+C$

(C) $2\ln|(x+3)|-3\ln|x+2|+C$

(D) $-2\ln|(x+3)|-3\ln|x+2|+C$

4. Let f be the function given by $f(\theta)=\int \frac{\sin\theta}{\cos\theta(\cos\theta-1)}\,d\theta$.

(a) Substitute $x=\cos\theta$ and write an integral expression for f in terms of x.

(b) Use the method of partial fractions to find $f(\theta)$.

6.6 Integration by Parts BC

Integration by Parts Formula

If u and v are functions of x and have continuous derivatives, then

$$\int u\,dv = uv - \int v\,du .$$

Guidelines for Integration by Parts

1. For integrals of the form

   $\int x^n e^{ax}\,dx$, $\int x^n \sin ax\,dx$, or $\int x^n \cos ax\,dx$

   let $u = x^n$ and let $dv = e^{ax}\,dx$, $\sin ax\,dx$, or $\cos ax\,dx$.

2. For integrals of the form

   $\int x^n \ln x\,dx$, $\int x^n \arcsin ax\,dx$, $\int x^n \arccos ax\,dx$, or $\int x^n \arctan ax\,dx$

   let $u = \ln x$, $\arcsin ax$, $\arccos ax$, or $\arctan ax$ and let $dv = x^n\,dx$.

3. For integrals of the form

   $\int e^{ax} \sin bx\,dx$ or $\int e^{ax} \cos bx\,dx$

   let $u = \sin bx$ or $\cos bx$ and let $dv = e^{ax}\,dx$.

Tabular Method

In problems involving repeated applications of integration by parts, a tabular method can help to organized the work. This method works well for integrals of the form $\int x^n e^{ax}\,dx$, $\int x^n \sin ax\,dx$, and $\int x^n \cos ax\,dx$.

Example 3 □ Evaluate $\int x^2 e^x\,dx$.

Solution □ Let $u = x^2$ and $dv = e^x\,dx$.

u and its Derivatives		v' and its Antiderivatives	
x^2	+	e^x	
$2x$	−	e^x	$+x^2e^x$
2	+	e^x	$-2xe^x$
0		e^x	$+2e^x$

Differentiate until you obtain 0 as aderivative.

Hence, $\int x^2 e^x\,dx = x^2e^x - 2xe^x + 2e^x + C$

Exercises

1. $\int x\sin(2x)\,dx =$

(A) $-x\cos(2x)+\frac{1}{2}\sin(2x)+C$

(B) $\frac{x}{2}\cos(2x)-\frac{1}{4}\sin(2x)+C$

(C) $-\frac{x}{2}\cos(2x)+\frac{1}{4}\sin(2x)+C$

(D) $\frac{x}{2}\cos(2x)+\frac{1}{4}\sin(2x)+C$

2. $\int_0^2 xe^x\,dx =$

(A) e^2-1 (B) e^2+1 (C) $e-1$ (D) $e+1$

3. If $\int x^2\cos(3x)\,dx = f(x)-\frac{2}{3}\int x\sin(3x)\,dx$, then $f(x) =$

(A) $\frac{2}{3}x\sin(3x)$

(B) $\frac{1}{3}x^2\sin(3x)$

(C) $\frac{2}{3}x\cos(3x)$

(D) $\frac{1}{3}x\sin(3x)-\frac{2}{3}\cos(3x)$

4. $\int_0^{\pi/4} x\sec^2 x\,dx =$

(A) $\frac{\pi}{4}-\ln 2$ (B) $\frac{\pi}{4}+\ln 2$ (C) $\frac{\pi}{4}-\frac{\ln 2}{2}$ (D) $\frac{\pi}{4}+\frac{\ln 2}{2}$

6.7 Improper Integrals BC

Improper Integrals with Infinite Integration Limits

1. If $f(x)$ is continuous on $[a,\infty)$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx .$$

2. If $f(x)$ is continuous on $(-\infty,b]$, then

$$\int_{-\infty}^b f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx .$$

3. If $f(x)$ is continuous on $(-\infty,\infty)$, then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^c f(x)\,dx + \int_c^{\infty} f(x)\,dx ,$$

where c is any real number.

Improper Integrals with Infinite Discontinuities

1. If $f(x)$ is continuous on $[a,b)$ and has an infinite discontinuity at b, then

$$\int_a^b f(x)\,dx = \lim_{c\to b^-}\int_a^c f(x)\,dx .$$

2. If $f(x)$ is continuous on $(a,b]$ and has an infinite discontinuity at a, then

$$\int_a^b f(x)\,dx = \lim_{c\to a^+}\int_c^b f(x)\,dx .$$

3. If $f(x)$ is continuous on $[a,b]$, except for some number c in (a,b) at which f has an infinite discontinuity, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx ,$$

where c is any real number.

In each case, if the limit is finite we say that the improper integral **converges** and that the limit is the value of the improper integral. If the limit fails to exists, the improper integral **diverges**.

Exercises

1. $\displaystyle\int_2^{\infty} \frac{1}{\sqrt{x-1}}\,dx =$

(A) $-\infty$ (B) -2 (C) 1 (D) ∞

2. $\displaystyle\int_0^{\infty} \frac{1}{(x+3)(x+4)}\,dx =$

(A) $-\ln\frac{4}{3}$ (B) $-\ln\frac{3}{4}$ (C) 0 (D) $\ln 4$

3. $\int_0^4 \frac{dx}{(x-1)^{2/3}} =$

(A) $3\sqrt[3]{3}$ (B) $3(1-\sqrt[3]{3})$ (C) $3(1+\sqrt[3]{3})$ (D) divergent

4. $\int_0^\infty x^2 e^{-x^3} =$

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) divergent

5. If $\int_0^1 \frac{ke^{-\sqrt{x}}}{\sqrt{x}}\,dx = 1$, what is the value of k ?

(A) $-\frac{1}{2}$ (B) $\frac{e}{2}$ (C) $\frac{1}{2}$ (D) There is no such value of k